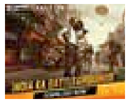
**Youtube web to redesign**
YouTube is testing description redesign for web with more focus on comments.
http://dgit.in/Jul21-65



**BGMI launched on PlayStore**
Battlegrounds Mobile India goes live on Play Store for Android Smartphones.
http://dgit.in/Jul21-66

```
dev/sda4 /mnt
for i in /dev/dev/pts/proc/
sys/run; do sudo mount -B $i
/mnt$i; done
sudo cp /etc/resolv.conf /
mnt/etc/
sudo chroot /mnt
```

**fstab**

First, we edit the fstab

fstab is the configuration file which contains all the partitions which should be automatically mounted on boot. We change this file manually via the nano text editor. We want to add additional boot options to the last line, and then add another line for the home partition. As the UUID of the home partition needs to be the same as that of /, we will copy-paste the edited last line, and simply add `@home` instead of `@`. We want the following boot options: `defaults,space_cache,noatime,compress=zstd,commit=120,ssd,discard,autodefrag,subvol=Z`

For SSDs, you should omit adding `autodefrag`. For HDDs, you should omit adding `ssd,discard`.

Open the file with nano

```
nano /etc/fstab
```

In the last line, find the word `defaults`, and replace it with `defaults,space_cache,noatime,compress=zstd,commit=120,ssd,discard,autodefrag,subvol=@`. Remember to change the line depending on if you have HDDs or SSDs. Once you have made the changes, hit the `End` key to transport the cursor at the end of the line. Press the key combination `Alt+A` to set a mark, and hit the `Home` key to mark the entire line. Then press the key combination `Alt+6` to copy the line. Go to the next line, and paste the line by pressing the combination `Ctrl+U`. Then edit the `/` to become `/home`, and `subvol=@` to become `subvol=@home`. Note that the UUID for both the root partition and the home partition must match. The file should look like the following:

```
PARTUUID=aaaaaaaa-aaaa-aaaa-
aaaa-aaaaaaaa  /boot/efi
vfat  umask=0077  0  0
PARTUUID=bbbbbbbb-bbbb-bbbb-
bbbb-bbbbbbbb /recovery vfat
umask=0077 0 0
/dev/mapper/cryptswap  none
swap  defaults  0  0
UUID=yyyyyyyy-yyyy-yyyy-
yyyy-yyyyyyyyyyyy  /  btrfs
defaults,space_cache,noatime,
compress=zstd,commit=120,ssd,
discard,subvol=@  0  0
UUID=yyyyyyyy-yyyy-yyyy-yyyy-
yyyyyyyyyyyy  /home  btrfs
defaults,space_cache,noatime,
compress=zstd,commit=120,ssd,
discard,subvol=@home  0  0
```

We can check that the fstab is right by running the mount command: sudo mount -av. If the output shows errors of any kind, your system will fail to boot, and you shuold start over, or seek help. The output should look something like as below:

```
  /boot/efi : successfully
mounted
  /recovery : successfully
mounted
  none : ignored
  / : ignored
  /home : successfully
mounted
```

**kernelstub**

We now add changes to the kernelstub configuration. Kernelstub is the bootloader, the program which actually loads the Linux kernel and starts your OS. We indicate to the loader the subvolume where our root filesystem resides. We also update the init file to be double sure that our changes have propogated to all the required places.

```
  kernelstub -a
'rootflags=subvol=@' -l -s
  update-initramfs -c -k all
```

**loader.conf**

This file manages the settings of the bootloader screen. We add a timeout of 5 seconds, so its easier to boot into the firmware or access recovery options. This is an optional change, so you can skip this if you don't want a 5 second delay each time you start your computer.

```
echo "timeout 5" >> /boot/
efi/loader/loader.conf
```

**Reboot**

If you have reached this point, and not run into any errors, you can now reboot!

Step out of the chroot environment by typing exit at the terminal, and then close the terminal. Go back to the Pop!_OS installer, and hit "Restart Device". You shouould now boot into a shiny new install of Pop!_OS, with a side of butter! The first time boot will welcome you to your computer, and help you set up your User Account.

You now have a fully functional system, but continue reading for quality of life configuration like backups and data deduplication.

## TIMESHIFT

Timeshift is a nifty tool which simplifies the backup process of your system. The utility is meant to roll back any breaking system changes by keeping versioned snapshots of your computer. It will save you from an accidental delete, or a weird update which breaks your environment. It will not save you from a failing disk on its own.

Note that snapshots on btrfs do not take much extra space, so you could have as many snapshots as required to fit into your backup strategy, as long as they're under a hundred. Its possible to have even more snapshots, but you'll start running into performance problems around then.

Also note that creating btrfs snapshots is an instant action that barely impacts system resources, but deleting snapshots is a bit more involved, and the system might feel a little sluggish for the few minutes it takes for btrfs to delete a snapshot.